

A set $F[x] = \{ \sum_{i=0}^n a_i \cdot x^i \mid n \in \mathbb{N}, a_i \in F \}$ with addition and scalar-multiplication is called the polynomial space. As we know, it is countably infinite dimensional.

$$F[x] = \text{span}_F \{ x^k \mid k \in \mathbb{N} \cup \{0\} \}.$$

In this paper, we investigate some typical properties of the polynomial space, using our background knowledge about dual, normal form, and inner product.



In $\mathbb{C}[x]$, the map

$$\langle \cdot, \cdot \rangle : \mathbb{C}[x] \times \mathbb{C}[x] \rightarrow \mathbb{C} : \int_I f(x) \cdot \overline{g(x)} dx \quad \text{where } I \text{ is closed interval}$$

is an inner Product on $\mathbb{C}[x]$, because: given $f, h, g \in \mathbb{C}[x]$, and $c \in \mathbb{C}$,

$$\int_I [cf+h] \cdot \overline{g} = c \int_I f \cdot \overline{g} + \int_I h \cdot \overline{g} \quad \text{by the linearity of integral}$$

And

$$\int_I f \cdot \overline{g} = \int_I \overline{f \cdot g} = \int_I \overline{f} \cdot g = \int_I g \cdot \overline{f}, \quad \text{because}$$

$$\int_I f = \int_I \operatorname{Re} f + i \int_I \operatorname{Im} f = \int_I \operatorname{Re} f - i \int_I \operatorname{Im} f = \int_I \operatorname{Re} f - i \operatorname{Im} f = \int_I \overline{f}.$$

Finally

$$\int_I f \cdot \overline{f} = \int_I \|f\|^2 \geq 0. \quad \text{So, shown. } \square$$

Def. Let $V = \mathbb{C}[z]$, the space of complex coefficient polynomials.

Define an inner product on V as:

$$\langle f, g \rangle = \int_0^1 f(w) \overline{g(w)} dw, \text{ where } \overline{g} \text{ is obtained by taking conjugate to coeff. } g.$$

1. V does not satisfy the Riesz-representation Theorem.

Proof. Let $z \in \mathbb{C}$ be fixed. Let $L: V \rightarrow \mathbb{C}: f \mapsto f(z)$ be the evaluation at z .

Clearly, L is a linear functional. Suppose that for some $g \in V$ s.t.

$$\langle f, g \rangle = L(f) = f(z) \text{ for all } f \in V.$$

Take $h(w) = w - z$. Then, for any $f \in V$,

$$\langle h \cdot f, g \rangle = \int_0^1 h(t) \cdot f(t) \cdot \overline{g(t)} dt = h(z) \cdot g(z) = 0.$$

Put $f = \overline{h}g$, then

$$\int_0^1 |h(w)|^2 |g(w)|^2 dw = 0, \text{ so } h(w)g(w) = 0, \text{ but } h(w) \neq 0, \text{ so } g \equiv 0.$$

This implies $L \equiv 0$, but it is contradiction.

2. Adjoint of a linear operator on V may not exist.

Finite subspace of $F[x]$.

Let $P_2(\mathbb{R}) = \text{span}_{\mathbb{R}}\{1, x, x^2\}$, equipped inner product

$$\langle f, g \rangle = \int_0^1 f(t) \cdot g(t) dt.$$

1). For fixed $a \in \mathbb{R}$, let $V_a: P_2(\mathbb{R}) \rightarrow \mathbb{R}; f \mapsto f(a)$.

Find a polynomial $h \in P_2(\mathbb{R})$ s.t.

$$\langle f, h \rangle = V_a(f) = f(a) \text{ for all } f \in P_2(\mathbb{R}).$$

Proof. Construct an orthonormal basis for $P_2(\mathbb{R})$ as:

$$\alpha_1 = 1.$$

$$\alpha_2 = x - \frac{\langle x, 1 \rangle}{\|1\|^2} \cdot 1 = x - \frac{1}{2}.$$

$$\alpha_3 = x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} \cdot 1 - \frac{\langle x^2, x - \frac{1}{2} \rangle}{\|x - \frac{1}{2}\|^2} \cdot (x - \frac{1}{2}) = x^2 - \frac{1}{3} - (x - \frac{1}{2}) = x^2 - x + \frac{1}{6}.$$

$$\langle x^2, x - \frac{1}{2} \rangle = \int_0^1 x^3 - \frac{1}{2}x^2 dx = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}.$$

$$\langle x - \frac{1}{2}, x - \frac{1}{2} \rangle = \int_0^1 (x - \frac{1}{2})^2 dx = 2 \cdot \int_0^{\frac{1}{2}} x^2 dx = 2 \cdot \frac{1}{3} \cdot \frac{1}{8} = \frac{1}{12}.$$

$$\begin{aligned} \langle x^2 - x + \frac{1}{6}, x^2 - x + \frac{1}{6} \rangle &= \int_0^1 x^4 - x^3 + \frac{1}{6}x^2 - x^3 + x^2 - \frac{1}{6}x + \frac{1}{6}x^2 - \frac{1}{6}x + \frac{1}{36} dx \\ &= \int_0^1 x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36} dx \\ &= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{8} + \frac{1}{36} = \frac{1}{180}. \end{aligned}$$

Therefore, $\{1, \sqrt{12}(x - \frac{1}{2}), \sqrt{180} \cdot (x^2 - x + \frac{1}{6})\}$ is an orthonormal basis.

$$\begin{aligned} \text{Now, } h(x) &= 1 \cdot 1 + \sqrt{12} \cdot (a - \frac{1}{2}) \cdot \sqrt{12} (x - \frac{1}{2}) + \sqrt{180} \cdot (a^2 - a + \frac{1}{6}) \cdot \sqrt{180} (x^2 - x + \frac{1}{6}) \\ &= 1 + (12a - 6)(x - \frac{1}{2}) + 180(a^2 - a + \frac{1}{6})(x^2 - x + \frac{1}{6}) \end{aligned}$$

is that desired.

